

10 Chapters · 200+ Questions · 3 Quiz Levels · Free Access · No Account Required

COURSE OVERVIEW — CHAPTER BY CHAPTER

1. **Gateway Architecture** — Traditional SCADA platforms charged per tag and per client seat. Ignition's unlimited licensing model changed the economics of the entire market. This chapter covers Gateway installation, the module system, web interface, and the backup/restore procedures that protect a running production system.
2. **Tag System** — Every piece of data in Ignition is a tag. OPC tags read from PLCs. Expression tags compute derived values. Memory tags hold state. UDTs create typed, reusable templates for equipment that appears dozens of times on a single plant floor. The tag system is the foundation everything else builds on.
3. **OPC-UA Integration** — Ignition connects to Siemens, Allen-Bradley, Modbus, and any OPC-UA compliant device through a single protocol. This chapter covers Ignition as both OPC-UA server and client, and the device connection configuration that takes a new PLC from unconfigured to live tags in under an hour.
4. **Perspective Module** — HMI running in any browser, on any device, without plugins. This chapter covers the component library, View/Page/Session architecture, and the binding system that connects every displayed value to a live tag without writing a single line of code for the common case.
5. **Scripting** — Python (Jython) scripting runs on tag events, on gateway events, on a schedule, and in response to user actions. This chapter covers the system library that gives scripts access to tags, databases, alarm pipelines, and messaging — automation logic without compiling a PLC program.
6. **Alarm Notification** — An alarm that reaches no one is equivalent to no alarm. This chapter covers alarm modes, notification pipelines, escalation rules, roster configuration, and email/SMS delivery — the infrastructure that ensures a fault reaches the right person at 2 a.m.
7. **Tag Historian** — The historian stores process data for trend analysis, regulatory compliance, and root cause investigation. This chapter covers storage providers, partitioning strategy, aggregation modes, and Store and Forward — the feature that preserves data through network interruptions without gaps in the historical record.
8. **Database Integration** — Named queries connect Ignition to any SQL database without embedding SQL in scripts. Transaction Groups push tag data to database tables on scan or on change. This chapter closes the loop from field sensor to management report.
9. **Security & Access** — Identity providers, role mapping, project-level permissions, and SSL/TLS configuration. The Ignition security model scales from a single-site deployment with five users to a multi-site enterprise with hundreds of engineers and operators.
10. **Field Lab** — Build a complete SCADA application from scratch: PLC connection via OPC-UA, tag configuration, Perspective dashboard with live trends and alarm displays, alarm notification pipeline, historian storage, and a management report generated from historical data.

Certification Relevance

- **Inductive University Certified** — Core, Advanced, and Gold exams are free, self-paced, and industry-recognized; this course provides the structured foundation those exams assume.
- **ISA CCST** — SCADA system configuration, historian, and alarm management are tested topics at CCST Levels II and III.
- **Ignition Integrator** — Required for Inductive Automation Certified Integrators; demonstrates full-stack project delivery capability.

Next Steps

1. Download the Ignition trial from [inductiveautomation.com](https://www.inductiveautomation.com) (free, resets every two hours, fully featured — sufficient for the entire course).
2. Complete Inductive University Core certification (free, online, approximately 20 hours of video and exercises).
3. Build the Field Lab application from scratch using your own trial instance as the capstone.
4. Pursue Gold certification for the Certified Integrator pathway and higher-value project work.